

Docker Volume Not Persisting Data SOP

Objective

Prevent data loss when container data does not persist after restart, recreate, or redeployment because of incorrect volume configuration.

Scope

Applicable to Docker-based services, containerized application deployments, DevOps teams, SRE teams, platform teams, and synthetic incident workflows across development, staging, and production-like environments.

Pre-requisites

1. Access to Docker host, Docker CLI, container runtime logs, and the target container or image name.
2. Permission to inspect containers, images, volumes, networks, resource limits, and compose files where applicable.
3. Access to application logs, deployment artifacts, Dockerfile, environment variables, and registry credentials.
4. Monitoring access for host CPU, memory, disk, container restart count, and container health checks.
5. Escalation contact for application, DevOps, security, registry, infrastructure, and platform teams.

Incident Metadata

Field	Value
Ticket ID	T012
Issue	Docker volume not persisting data
Category	Storage
Service	Docker
Severity	Medium
Status	Resolved
Description	Data lost after container restart
Expected Resolution	Use named volumes instead of anonymous volumes

Related Log Evidence

Log ID	Timestamp	Level	Message	Error Code
N/A	N/A	N/A	No direct log evidence found in logs CSV for this ticket.	N/A

Procedure

1. Confirm the affected container, expected persistent path, data type, and when the data was lost.
2. Inspect container mounts to verify whether a named volume, bind mount, or anonymous volume is used.
3. Check Docker run or compose configuration for the correct volume mapping.
4. Validate that the application writes data to the mounted path and not to a temporary internal path.
5. Inspect existing volumes and confirm data exists in the intended volume.
6. Replace anonymous volumes with named volumes or approved bind mounts.
7. Recreate the container carefully without deleting the persistent volume.
8. Validate data remains available after container restart and recreate.

Common Commands

```
docker ps -a
docker logs <container-name-or-id> --tail=100
docker inspect <container-name-or-id>
docker stats --no-stream
docker images
docker network ls && docker network inspect <network-name>
docker volume ls && docker volume inspect <volume-name>
docker system df
```

Troubleshooting Matrix

Issue	Possible Cause	Resolution
Data lost after restart	Anonymous volume recreated	Use named volume
Data not visible on host	Wrong mount path	Correct source and target mapping
Permission denied on volume	UID/GID mismatch	Fix ownership or container user
Old data not loaded	New volume attached	Attach correct existing volume

Best Practices

1. Use named volumes for stateful container data.
2. Avoid storing critical data only in container writable layer.
3. Back up production volumes before migrations.
4. Document mount paths in deployment runbooks.

References

Docker Docs | Internal Tickets CSV | Internal Logs CSV | Container Runtime Logs | Monitoring Dashboards | DevOps Runbook